

Scalability & Future Plan

Date	09 July 2026
Team ID	
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-threaded development server	Not suitable for concurrent production traffic	High
2	Local flat-file storage (CSV/Pickle)	Limited concurrency and scalability	Medium
3	Basic input validation	Potential for unhandled out-of-bounds errors	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Limited local concurrency	Deploy with Gunicorn/Nginx or cloud hosting (AWS/GCP)
2	Data Storage	Local CSV files	Transition to a relational database (e.g., PostgreSQL/SQLAlchemy)
3	Performance	Synchronous processing	Implement task queues (e.g., Celery) and caching (e.g., Redis)
4	Security	Open access	Implement authentication (e.g., OAuth/JWT) and enhanced input sanitization

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	User Authentication & History Tracking	Q3 2026	Enhanced user personalization and data persistence
Phase 3	Real-time API Data Integration	Q4 2026	Automated updates via World Bank/UN API feeds
Phase 4	Advanced Model Migration (Neural Networks)	Q1 2027	Improved prediction accuracy and feature interaction mapping